

Function and description

```
bit[len]
Crypto_DRBG(int len)
```

Returns an array of **len** random bits.

Mapping (Version 1.0)

Crypto_DRBG() SHALL be implemented with one of the following DRBG algorithms as defined in [NIST 800-90A](#) (the choice of which one is left to the manufacturer because the choice has no impact on the interoperability):

- CTR DRBG (with AES-CTR)
- HMAC DRBG (with SHA-256)
- HMAC DRBG (with SHA-512)
- Hash DRBG (with SHA-256)
- Hash DRBG (with SHA-512)

Crypto_DRBG() SHALL be seeded and reseeded using **Crypto_TRNG()** with at least 256 bits of entropy (see among others Chapter 4, Section 8.4, and Section 8.6.8 of [NIST 800-90A](#)).

3.2. True Random Number Generator (TRNG)

A TRNG (a.k.a. Entropy Source) is required to provide an entropy seed as an input to the DRBG algorithm.

Function and description

```
bit[len]
Crypto_TRNG(int len)
```

Returns an array of **len** random bits.

Mapping (Version 1.0)

Crypto_TRNG() MAY be implemented according to the [NIST 800-90B](#) implementation guidelines but alternate implementations MAY be used.

In accordance with good security practices, the **Crypto_TRNG()** SHALL never be called directly but rather SHALL be used in the implementation of **Crypto_DRBG()**.

3.3. Hash function (Hash)

Crypto_Hash() computes the cryptographic hash of a message.